Project Assignment

Student Performance Prediction and Academic Risk Analysis

1. Problem Statement

An academic department wants to identify students who may need additional academic support. You are given student study habits, attendance, scores, previous GPA, and learning support information. Your task is to analyze the data, predict final performance, classify academic risk level, and suggest simple support strategies.

2. Dataset Description

Use the provided dataset file: student_performance_project_dataset.csv

The dataset contains 700 student records with the following columns:

Column(s)	Meaning
StudentID	Unique student identifier
Gender	Student gender
StudyHoursPerWeek	Average weekly study time
AttendancePercent	Class attendance percentage
AssignmentScore, QuizAverage, ProjectScore	Academic performance features
PreviousGPA	Previous academic performance
SleepHours	Average sleep duration
InternetAccess, ExtraClasses, ParentEducation	Support and background features
FinalScore	Final numerical score
PassStatus	Pass or fail label
RiskLevel	Low Risk, Medium Risk, or High Risk label

3. Colab Setup Instructions

1. Open Google Colab and create a new notebook.
2. Upload student_performance_project_dataset.csv using the Files panel or files.upload().
3. Import Pandas, NumPy, Matplotlib, and required scikit-learn libraries.
4. Read the CSV file and display the first few rows before starting the analysis.

```
from google.colab import files
uploaded = files.upload()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('student_performance_project_dataset.csv')
df.head()
```

4. Tasks

Part A: Data Loading and Understanding

5. Load the dataset and display the first 10 rows.
6. Print dataset shape, column names, data types, and missing value count.
7. Find summary statistics for all numerical columns.
8. Count students in each PassStatus and RiskLevel category.
9. Identify which features appear most related to FinalScore.

Part B: Probability and Basic Analysis

10. Find the probability that a randomly selected student passes.
11. Find the probability that a student is in the High Risk category.
12. Find the probability that AttendancePercent is greater than 75%.
13. Find the probability that StudyHoursPerWeek is above the dataset mean.
14. Compare pass probability for students with and without ExtraClasses.

Part C: Exploratory Data Analysis and Visualization

15. Plot histograms for StudyHoursPerWeek, AttendancePercent, PreviousGPA, and FinalScore.
16. Create a scatter plot of StudyHoursPerWeek vs FinalScore.
17. Create a scatter plot of AttendancePercent vs FinalScore.
18. Create a box plot of FinalScore grouped by RiskLevel.
19. Create a correlation heatmap and write 5 observations.

Part D: Data Preprocessing

20. Separate numerical and categorical columns.
21. Encode categorical features such as Gender, InternetAccess, ExtraClasses, ParentEducation, PassStatus, and RiskLevel.
22. Scale numerical features using StandardScaler.
23. Create two target variables: FinalScore for regression and RiskLevel or PassStatus for classification.
24. Split data into training and testing sets.

Part E: Regression Task - Predict Final Score

25. Train at least three regression models: Linear Regression, Decision Tree Regressor, and Random Forest Regressor or KNN Regressor.
26. Evaluate using MAE, MSE, RMSE, and R2 score.
27. Plot actual vs predicted FinalScore for the best model.
28. Identify which features help predict FinalScore.
29. Write a short explanation of model errors.

Part F: Classification Task - Predict Risk Level or Pass Status

30. Train at least three classification models: Logistic Regression, Decision Tree, and KNN or Random Forest.
31. Evaluate using accuracy, confusion matrix, precision, recall, and F1-score.
32. Compare models in a table.
33. Explain which model is best for identifying students who need support.
34. Discuss whether accuracy alone is enough for this problem.

Part G: Gradient Descent or Neural Network

35. Implement simple gradient descent for linear regression using NumPy, or train an MLPClassifier/MLPRegressor using scikit-learn.
36. Plot the loss curve or training performance curve.
37. Compare the neural network or gradient-descent result with one traditional model.
38. Explain why learning rate and iterations affect model performance.

Part H: Dimensionality Reduction

39. Apply PCA and reduce data to 2 components.
40. Plot PCA points colored by RiskLevel.
41. Apply LDA for supervised dimensionality reduction if using RiskLevel as target.
42. Apply t-SNE and compare with PCA.
43. Explain which method separates risk groups better.